FIZZAH AMIR

Backend Developer | Django, PostgreSQL, REST APIs

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PROFILE

Third-year Computer Science undergraduate at Information Technology University (ITU) with hands-on experience building and deploying backend systems: a 29-endpoint Django REST API with PostgreSQL triggers, explicit transactions, JWT authentication and role-based access control, live on AWS EC2, plus a full-stack RAG document Q&A application. Strong in SQL, database design and API security. Seeking a backend internship to build reliable, well-structured services within a team.

EDUCATION

BS Computer Science | Information Technology University (ITU), Lahore 2024 – 2028

Coursework: Data Structures and Algorithms, Advanced Database Management, Object-Oriented Programming, Network Programming

FSc, Pre-Engineering | Kinnaird College for Women, Lahore 2022 – 2024

TECHNICAL SKILLS

Backend: Python, Django, Django REST Framework, REST APIs, JWT Authentication, Argon2, Role-Based Access Control, C++
Databases: PostgreSQL, Oracle Database, SQL (joins, triggers, transactions, parameterized queries), Schema Design
Tools & Cloud: Git, GitHub, Linux, AWS EC2, Railway Cloud, React (basic)
AI / Data: RAG, FAISS, Sentence Transformers, Scikit-learn, TensorFlow, XGBoost, Pandas, NumPy, Matplotlib

CERTIFICATIONS

- **Meta Backend Developer Professional Certificate**, Meta (Coursera)
- Google (Coursera): Foundations of Data Science (Jun 2026), The Power of Statistics, Go Beyond the Numbers
- Cybersecurity Bootcamp: ISO 27001:2022 (Apr 2025)

PROJECTS

Digital Wallet System API Live on AWS EC2

- Designed a relational PostgreSQL database and 29 REST API endpoints in Django REST Framework using raw SQL, explicit transactions and parameterized queries.
- Wrote 8 PostgreSQL triggers and a 5-level automatic rank system; implemented JWT authentication, Argon2 password hashing and Role-Based Access Control across 3 roles.

Tech: Python, Django REST Framework, PostgreSQL, SQL, AWS EC2

Insightly-PDF: RAG-Based Document Q&A Assistant Personal Project

- Built a full-stack application that extracts and chunks text from uploaded PDFs and returns citation-backed answers to natural-language questions.
- Developed the Django REST backend for ingestion, embedding generation and vector-search retrieval, with a React chat-style frontend.

Tech: Python, Django REST Framework, React, Vector Search, RAG, REST APIs

In-Memory Cache Server (Redis Clone) ITU | Course Project

- Built a single-threaded key-value server in C++ implementing the Redis protocol (RESP) with crash-safe persistence; sustained 50,000+ SET/GET operations per second in benchmarks.

Tech: C++, epoll, TCP Sockets, Skip Lists, Hash Tables

AeroHub Management System ITU | Dec 2025 – Jan 2026

- Built an airport management system in C++ with a custom HTTP server deployed on Railway Cloud; implemented B-Tree, Dijkstra's algorithm, HashMap and Priority Queue from scratch.

Tech: C++, REST API, HTML/CSS/JS, Railway Cloud

Electricity Theft Detection Using Machine Learning ITU | Group Project

- Analyzed 2M+ smart-meter records (UCI dataset); cleaned data, engineered consumption features and benchmarked Logistic Regression, Decision Tree, Random Forest, XGBoost and a TensorFlow neural network, using SMOTE for class imbalance.

Tech: Python, Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow

Meeting Transcriber and Q&A Assistant (RAG) Personal Project

- Transcribed audio, chunked transcripts, embedded them with Sentence Transformers (all-MiniLM-L6-v2) and indexed in FAISS to answer queries with grounded responses; added summarization and action-item extraction.

Tech: Python, FAISS, Sentence Transformers, RAG, Speech-to-Text

EXPERIENCE

Webmaster, Core Team Member | IEEE ITU Student Branch Aug 2026 – Present

- Manage and maintain the society's web presence; built a Python chatbot for IEEE Orientation that answers student queries in real time.

Campus Ambassador | Comeback Pakistan, ITU Sep 2026 – Present

- Represent Comeback Pakistan's AI Accelerator Program at Information Technology University (ITU).

ACHIEVEMENTS

- **Alibaba Hackathon:** qualified for the regional round in the Open Innovation category with **SOS**, an emergency app.
- **CodeRush, ITU (May 2025):** participated in the competitive programming contest, solving algorithm and data structure problems under time constraints.
- **SOFTEC 2026:** participated in the competitive programming competition.